Program: Maven Ansible Webapp

Step 1: Create Project

mvn archetype:generate -DgroupId=com.example -DartifactId=AnsibleApp -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false

```
harith@harith-VirtualBox:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=AnsibleApp -DarchetypeArtifactId=
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 10 MB/s)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/harith/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: AnsibleApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/harith/devops/AnsibleApp
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.468 s
[INFO] Finished at: 2026-05-11T05:51:24+05:30
[INFO] ------------------------------------------------------------------------
harith@harith-VirtualBox:~/devops$
```

Step 2: Edit Pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/maven-
v4_0_0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.example</groupId>
<artifactId>MavenAnsibleWebApp1</artifactId>
<packaging>war</packaging>
<version>1.0-SNAPSHOT</version>
<name>MavenAnsibleWebApp1</name>
<url>http://maven.apache.org</url>
<dependencies>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>3.8.1</version>
<scope>test</scope>
</dependency>
<!-- Servlet API for Java EE -->
```

```
<dependency>
<groupId>javax.servlet</groupId>
<artifactId>javax.servlet-api</artifactId>
<version>4.0.1</version>
<scope>provided</scope>
</dependency>
</dependencies>
<build>
<plugins>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-war-plugin</artifactId>
<version>3.4.0</version>
<!-- Update to latest stable version -->
</plugin>
</plugins>
<finalName>MavenAnsibleWebApp1</finalName>
<!-- create a web page with the name MavenAnsibleWebApp, if we want some other
name we
can chage it -->
</build>
</project>
```

Step 3: Create Ansible Configuration Files

In this step, we create a new folder named ansible inside the project directory using mkdir ansible.
Then, we move into the folder using cd ansible and create the required Ansible files using gedit
hosts.ini and gedit playbook.yml. These files are used to define target hosts and deployment tasks
for automation.

```
harith@harith-VirtualBox:~/devops$ cd AnsibleApp/
harith@harith-VirtualBox:~/devops/AnsibleApp$ ls
pom.xml  src
harith@harith-VirtualBox:~/devops/AnsibleApp$ mkdir ansible
harith@harith-VirtualBox:~/devops/AnsibleApp$ gedit hosts.in
harith@harith-VirtualBox:~/devops/AnsibleApp$ ge
geckodriver  genisoimage  geqn         getconf      getent       getkeycodes  getopts      gettext      getty
gedit        genl         getcap       geteltorito  getfacl      getopt       getpcaps     gettext.sh   getweb
harith@harith-VirtualBox:~/devops/AnsibleApp$ gedit playbook.yml
harith@harith-VirtualBox:~/devops/AnsibleApp$ ls
ansible  hosts.in  playbook.yml  pom.xml  src
harith@harith-VirtualBox:~/devops/AnsibleApp$
```

gedit host.ini
This command is written inside hosts.ini file to define the Ansible inventory. It creates a group
named web and adds localhost as the target system, using local connection and Python 3
interpreter for executing Ansible tasks.

localhost ansible_connection=local ansible_python_interpreter=/usr/bin/python3

gedit playbook.yml
This code is written inside playbook.yml file to automate deployment using Ansible. It targets the web host group, copies the generated WAR file from the Jenkins workspace to the Tomcat webapps folder, and then restarts the Tomcat service so the updated application runs successfully.

- name: Deploy to Tomcat
hosts: web
become: yes

tasks:
- name: Copy WAR file to Tomcat
copy:
src: "/var/snap/jenkins/5013/workspace/MavenAnsibleWebApp1-CICD/target/MavenAnsibleWebApp1.war"
dest: "/opt/tomcat/webapps/MavenAnsibleWebApp1.war"
remote_src: no

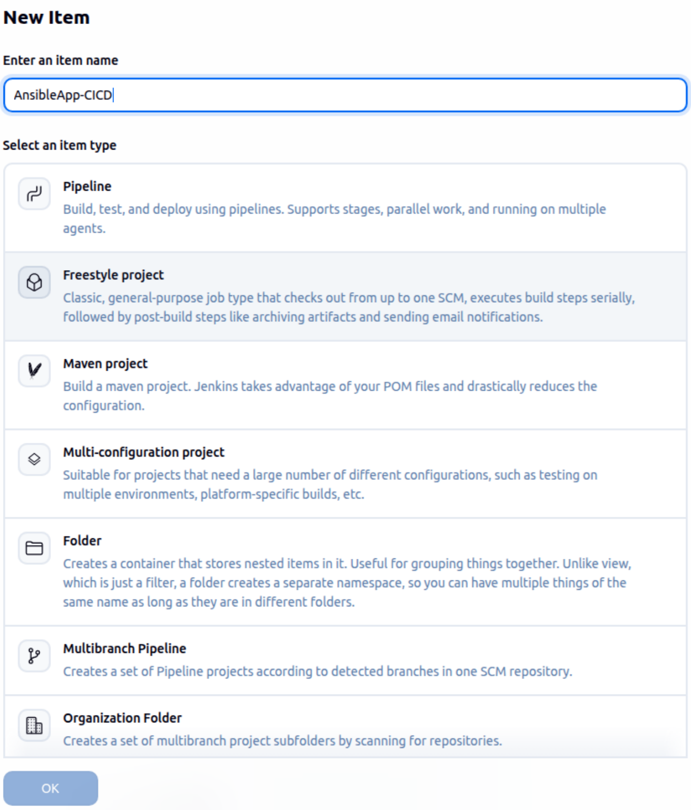
- name: Restart Tomcat
systemd:
name: tomcat
state: restarted

Step 4: Build the project

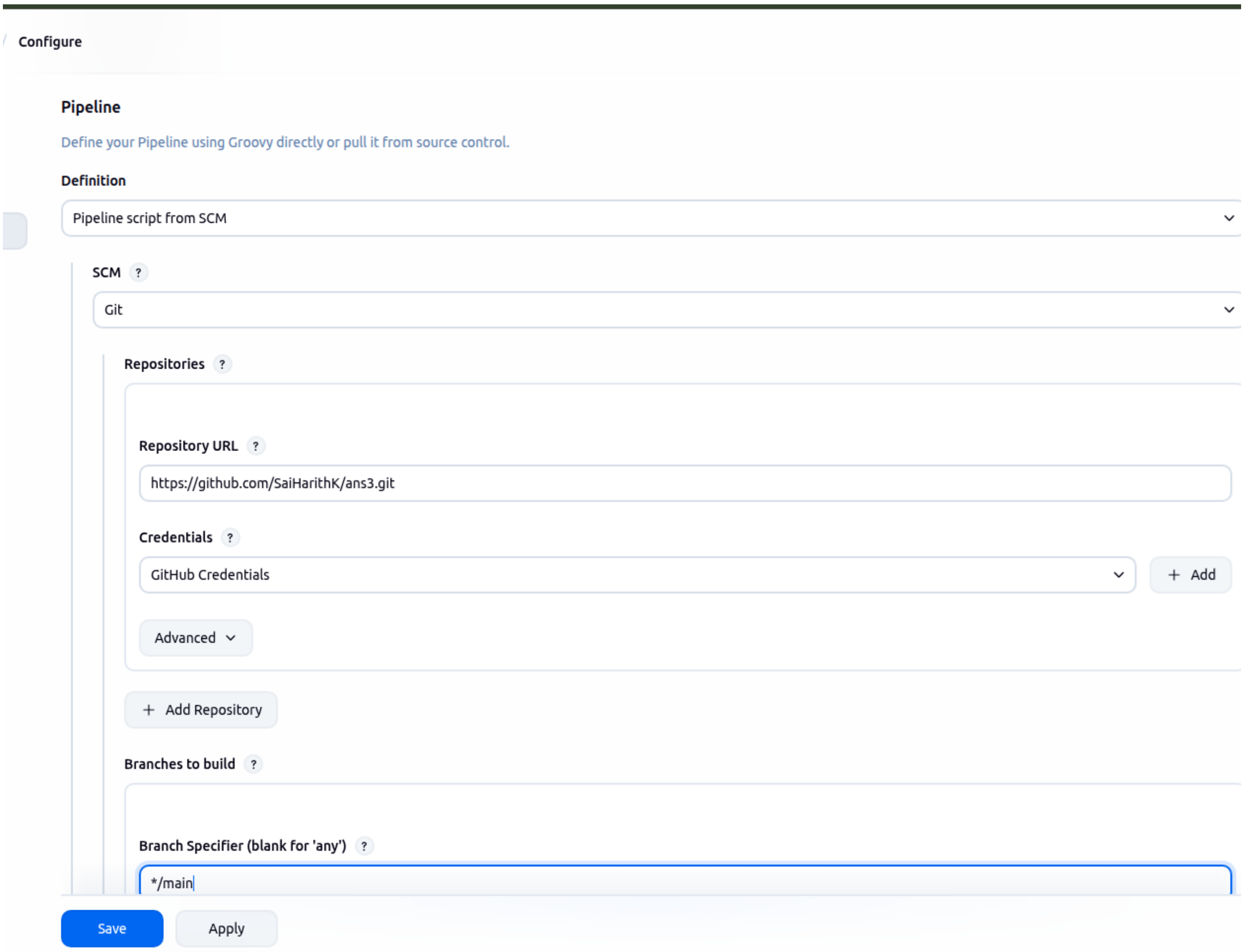
In this step, we run the command mvn clean install to clean previous files and build the project. Maven compiles the application, packages it as a .war file, and stores it in the target folder.

```
harith@harith-VirtualBox:~/devops$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< com.example:AnsibleApp >-----------------------
[INFO] Building AnsibleApp 1.0-SNAPSHOT
[INFO] --------------------------------[ war ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ AnsibleApp ---
[INFO] Deleting /home/harith/devops/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ AnsibleApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/harith/devops/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ AnsibleApp ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ AnsibleApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/harith/devops/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ AnsibleApp ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ AnsibleApp ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.4.0:war (default-war) @ AnsibleApp ---
[INFO] Packaging webapp
[INFO] Assembling webapp [AnsibleApp] in [/home/harith/devops/target/AnsibleApp]
[INFO] Processing war project
[INFO] Building war: /home/harith/devops/target/AnsibleApp.war
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ AnsibleApp ---
[INFO] Installing /home/harith/devops/target/AnsibleApp.war to /home/harith/.m2/repository/com/example/AnsibleApp/1.0-SNAPSHOT/
[INFO] Installing /home/harith/devops/pom.xml to /home/harith/.m2/repository/com/example/AnsibleApp/1.0-SNAPSHOT/AnsibleApp-1.0
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.366 s
[INFO] Finished at: 2026-05-11T06:15:44+05:30
[INFO] ------------------------------------------------------------------------
```

Step 5: Jenkins Pipeline



Step 6: Select pipeline Script form SCM



Step 7: Build Now